

Practical No. 14

Name : Danish Altaf Solkar

Enrolment no. : 24111590805

Class : CO - B

Batch : 2

Aim : Calculate cost of the project using COCOMO (Constructive Cost Model) / COCOMO II Approach

Aim

To estimate the effort, development time, and approximate project cost of an Online Food Ordering System using the Basic COCOMO (Constructive Cost Model) approach.

Theory

COCOMO is a software cost estimation model developed by Barry W. Boehm. It estimates the effort and development time required for a software project primarily from its estimated size in KLOC (thousand lines of code).

Basic COCOMO equations:

Effort (E) = $a \times (\text{KLOC})^b$ person-months

Development Time (T) = $c \times (E)^d$ months

Average Staff = E / T persons

Project Cost = $E \times \text{Cost per Person-Month}$

COCOMO Project Modes

Mode	Typical Project	a	b	c	d
Organic	Small/simple projects	2.4	1.05	2.5	0.38
Semi-detached	Medium complexity projects	3.0	1.12	2.5	0.35
Embedded	Complex, tightly constrained projects	3.6	1.20	2.5	0.32

Given Data / Assumptions

Parameter	Assumed Value
Project	Online Food Ordering System
Estimated size	20 KLOC
Project mode	Organic
Cost per person-month	■50,000

Calculation

Step 1 – Select COCOMO mode

The Online Food Ordering System is considered a relatively small and familiar business application. Therefore, the Organic mode is selected.

Step 2 – Calculate Effort

$$E = 2.4 \times (20)^{1.05}$$

$$E \approx 55.30 \text{ person-months}$$

Step 3 – Calculate Development Time

$$T = 2.5 \times (55.30)^{0.38}$$

$$T \approx 12.25 \text{ months}$$

Step 4 – Calculate Average Team Size

$$\text{Average Staff} = 55.30 / 12.25 \approx 4.51 \text{ persons}$$

Therefore, approximately 5 team members would be required on average.

Step 5 – Calculate Project Cost

$$\text{Cost} = 55.30 \times \text{₹}50,000$$

$$\text{Cost} \approx \text{₹}27,65,000$$

Result

Parameter	Estimated Value
Software size	20 KLOC
COCOMO mode	Organic
Effort	≈ 55.30 person-months
Development time	≈ 12.25 months
Average team size	≈ 4.51 persons (about 5)
Cost per person-month	₹ 50,000
Estimated project cost	≈ ₹ 27,65,000

Conclusion

Using the Basic COCOMO model, the estimated 20 KLOC Online Food Ordering System requires approximately 55.30 person-months of effort and 12.25 months of development time. With an assumed cost of ₹50,000 per person-month, the estimated project cost is approximately ₹27.65 lakh.